Honors Geometry – 3.5 Practice Name: _____ Period: _____

Section 3.5 — Equations of Parallel and Perpendicular Lines

Part A — Slope first. Everything in 3.5 starts with a slope. Get these right and the rest is bookkeeping.

Worked for you — follow these two steps every time. Find the slope through (1, −2) and (5, 6).

Step 1 — label the points: $x_1 = 1$, $y_1 = -2$, $x_2 = 5$, $y_2 = 6$

Step 2 — subtract in the SAME order, top and bottom: $m = (6 - (-2)) / (5 - 1) = 8/4 = 2$

1. Find the slope of the line through (−3, 8) and (5, −4). m = (___ − ___) / (___ − ___) = _____ answer: _____	**2. Find the slope of the line through (2, −7) and (2, 5).** m = (___ − ___) / (___ − ___) = _____ answer: _____ ★ **Think deeper:** Before you calculate: how could you tell from the two points alone that this slope would be undefined? _____
3. Find the slope of the line through (−6, 4) and (3, 4). m = (___ − ___) / (___ − ___) = _____ answer: _____	**4. A line with slope −2/5 passes through (−1, 6) and (9, ___). Find the missing coordinate.** −2/5 = (___ − 6) / (9 − (−1)) answer: _____

✓ **Check yourself:** Your four answers should be, in some order: one negative fraction, one undefined, one zero, and one positive whole number. If you have two of anything, one of them is wrong — #2 and #3 are the usual suspects.

Part B — Parallel, perpendicular, coinciding, or neither? Rewrite anything that is not already solved for y. Compare slopes, THEN compare y-intercepts.

The decision, in order.

Same slope, different b → parallel. Same slope, same b → coinciding (one line wearing two outfits). Slopes multiply to −1 → perpendicular.

Anything else → neither. A vertical line and a horizontal line are always perpendicular.

1. $y = \frac{3}{4}x - 2$ and $y = \frac{3}{4}x + 5$ m_1 = ___ m_2 = ___ → _____ answer: _____	**2. $y = -5x + 1$ and $x - 5y = 15$** rewrite: y = _____ answer: _____	**3. $2x + 3y = 12$ and $y = -\frac{2}{3}x + 4$** rewrite: y = _____ answer: _____ ★ **Think deeper:** Coinciding lines have equal slopes. Why don't we just call them parallel? _____
4. $y = 2x + 7$ and $y = -2x + 7$ m_1 = ___ m_2 = ___ → _____ answer: _____	**5. $x = -4$ and $y = 9$** vertical? horizontal? _____ answer: _____	**6. $6x - 2y = 8$ and $y = 3x + 1$** rewrite: y = _____ answer: _____

✓ **Check yourself:** You should have exactly two parallel, two perpendicular, one coinciding, and one neither. If your tally is off, re-read whichever problem you did NOT rewrite into y = mx + b first.

Part C — Write the equation. Three moves every time: get the slope of the given line, adjust it (same for parallel, negative reciprocal for perpendicular), then push it through the point.

Worked for you — write the line through (3, 1) parallel to y = 2x − 9.

Step 1 — slope of the given line: m = 2. Step 2 — parallel, so the new slope is also 2.
Step 3 — point-slope, then solve for y: y − 1 = 2(x − 3) → y − 1 = 2x − 6 → y = 2x − 5
Step 4 — check: does (3, 1) satisfy it? 2(3) − 5 = 1 ✓

1. Through (4, −3), parallel to y = ½x + 6. answer: _____	**2. Through (−6, 1), perpendicular to y = 3x − 8.** answer: _____	**3. Through (2, 9), parallel to 4x + y = 5.** answer: _____
4. Through (−3, −2), perpendicular to 2x − 5y = 10. answer: _____	**5. Through (0, −7), parallel to y = −x + 4.** answer: _____ ★ **Think deeper:** Why didn't you need point-slope form for this one? _____	**6. Through (5, 1), perpendicular to y = ⅘x − 3.** answer: _____

Answer Bank

y = −4x + 17 **(O)**	y = ½x + 5 **(T)**	y = ½x − 5 **(S)**
y = −x − 7 **(E)**	y = −3x − 1 **(A)**	y = −5/4 x + 29/4 **(S)**
y = −⅓x − 1 **(L)**	y = −4x + 9 **(R)**	y = −5/2 x − 19/2 **(P)**

✓ **Check yourself:** Find each of your six answers in the Answer Bank and write its letter in the boxes below, in order 1 through 6. They spell one word, a word this whole section is about. Three bank entries are decoys: if you land on one, that problem is wrong and you know exactly which.

1	2	3	4	5	6

Part D — Push further. Same three moves, hiding behind a sentence.

1. Error analysis. Jo says the line through (1, 4) perpendicular to y = 2x + 3 is y = 2x + 2. Jo is wrong. Name the mistake in one sentence, then write the correct equation. answer: _____	**2. Write the equation of the line through (−2, 7) that is parallel to the x-axis. Then write the line through (−2, 7) that is perpendicular to the x-axis.** answer: _____
3. Line a passes through (−1, 2) and (3, k). Line a is perpendicular to y = −2x + 1. Find k. answer: _____ ★ **Think deeper:** What would k be if line a were PARALLEL to y = −2x + 1 instead? _____	**4. Line m is y = ⅓x + 2. Line n is perpendicular to m and passes through (6, −1). Do m and n cross on the y-axis? Find where they cross and justify your answer.** answer: _____

✓ **Check yourself:** Add your k from #3 to the x-coordinate of the crossing point in #4. You should get 8.5. If you do not, exactly one of those two is wrong, and #3 is the faster one to redo.

Finished early? Go back to any problem where you wrote the answer but not the work, and write the work. That's the half of a test that earns partial credit.